



274 lines (189 loc) · 8.22 KB



Preview

Code

Blame

Raw



Xindi Tao 06-10-2026

Today's goal

1. I broke down our previous discussions into four projects. They are meant to be specific enough to turn into small research papers, or combined into a bigger one.
 - i. Project 1: I'm close to finishing a first draft, but I don't have a conference/journal in mind (NICE?).
 - ii. Project 2: **Aim for NICE 2027**, deadline Nov 11th. Topic: Bio-Inspired Sensing → Event-Driven Sensing. Deadline approaching, this should be prioritized.
 - iii. Project 3: **Aim for ECC 2027**, deadline Oct 31st. Topic: Biological systems. Deadline approaching, should be prioritized.
 - iv. Project 4: Combine Projects 2, 3, and Tony's work, design an event-based CPG controller. **Aim for a journal**
2. I went deep into the first project: E coli motor movement. I wrote arguments for the Introduction (research gap, research question, significance), a working model (explained in Methods and tested in Results). **I want to know whether I'm on the right track to turn this into a paper, and what is missing for a paper.**
3. Projects 2-4 are empty: didn't have time😂 I want to write out my ideas and have it for this Thursday.
4. I pushed back writing the FYR supplements. I want to know where do I receive the decision to resubmit, and when to do so.
5. I might not have time for supervision for this semester, can do next semester. (This year I'm doing 3F1 demonstration)

Projects

- E coli flagella movement resembles a spike
- Spiking proprioception
- Spiking motor neuron activity
- Spike-based CPG controller

1. E coli flagella movement resembles a spike

◆ Introduction

Research question

The question is whether E coli's motor signal resembles that of a spike.

Research gap

We ask this question because a spike is prevalent in living organism motion, both single cell (paramecium) and multi-cellular. (Spikes are discrete signals over continuous time.) However, because the signals are of different types (electrical vs chemical), the analogy between E coli signaling and a spiking neuron are mostly descriptive; a quantitative comparison between the two is missing.

To make the comparison, we categorize both into the class "event", and emphasize its property of 'start and finish' and maybe robustness. We distinguish the two by comparing their time-scales and thresholds.

Importance

An event is a core concept to neuromorphic compute. We aim to add one more example to the 'event' concept, and emphasize slow chemical signals, not just spikes, are events. In the future an event can be defined at faster/slower timescales, composed of multiple events. All these events have patterns to them. We are looking for the patterns.

◆ Methods

Models available

Tu, ..., Berg, 2007: Maps Aspartate concentration to CheY-P (phosphorylated CheY) activity. But CheY-P to flagella state (CW, or CCW) is missing.

Alon 2019: Used in 4G7. Cites (Tu 2007), but it is simplified.

Vladimirov, ..., Sourjik, 2008: Maps Aspartate concentration to CheY-P, then maps CheY-P to motor. But CheY-P is not dynamical (not a state variable) and is simplified. This simplification makes pulse response inaccurate in shape.

Connection: The authors know each other. Victor Sourjik and worked with Howard Berg. Uri Alon worked with Stanislas Leibler. I infer their models are similar.

Difference: The three models have the same narrative: take in attractant concentration ([Asp]), output motor activity or its indicator (CCW bias or CheY-P activity). All take into account the slow receptor adaptation, which is good. (Vlad 2008) didn't care about turning CheY-P into a dynamic variable; it did so intentionally so that it can save time running the simulation (thus the model name 'RapidCell') - I want CheY-P to be a state variable, apart from that it's good.

(Vlad 2008) and (Tu 2007) have different equations and parameter values (TODO: compare in a chart). I gathered my version based on the following: (1) use one paper for most of the model (Vlad 2008); (2) change equations only when it's necessary; (3) use simple over complex equations.

My model

TODO: check the math, it's Codex

Part I (source: Tu 2007) $L(t), m(t) \rightarrow f_t(L, m) \rightarrow a(t) \rightarrow Y(t)$

- $L(t)$: α -methyl-DL-aspartate (mM)
- $m(t)$: Mean receptor methylation
- $f_t(L, m)$: Total free-energy difference of an effective receptor cluster
- $a(t)$: Mean receptor-cluster & CheA kinase activity, $\in [0, 1]$
- $Y(t)$: Intracellular CheY-P concentration

$$f_t(m, L) = N [f_m(m) + f_L(L)],$$

where

$$f_m(m) = \alpha(m_0 - m),$$

$$f_L(L) = \ln \left(1 + \frac{L}{K_I} \right) - \ln \left(1 + \frac{L}{K_A} \right),$$

- K_I, K_A : ligand dissociation constants of the inactive and active receptor conformations (I have no idea what this means).

$$a = G(f_t) = \frac{1}{1 + \exp(f_t)}$$

$$\frac{dY}{dt} = k_a a - \frac{Y}{\tau_Z}$$

- k_a : Effective CheA-to-CheY phosphorylation rate
- $\tau_Z = 0.5$: CheZ-controlled dephosphorylation time

Part II: feedback (source Tu et al., 2007) **This is incomplete, no** $\tilde{F}(a)$, N , a_0 , s_1 , α $\dot{m}(t) \rightarrow \dot{m}(t) = F(a) = r_m \tilde{F}(a)$, where $r_m = \frac{1}{\tau_m \alpha N a_0 (1-a_0)(-s_1)}$

- $\tau_m = 3$

Part III (source: Vlad et al., 2008) $Y(t) \rightarrow B_{\text{CCW}}(t)$

- $B_{\text{CCW}}(t)$: Counter-clockwise motor bias, $\in [0, 1]$

$$Y_{\text{rel}}(t) = \frac{\max[Y(t), 0]}{Y_0}$$

- $Y_0 = 3.1 \mu\text{M}$

$$B_{\text{CCW}}(t) = \left[1 + \left(\frac{1}{B_0} - 1 \right) Y_{\text{rel}}(t)^H \right]^{-1},$$

- $B_0 = 0.65$: adapted CCW bias
- $H = 10.3$: motor Hill coefficient

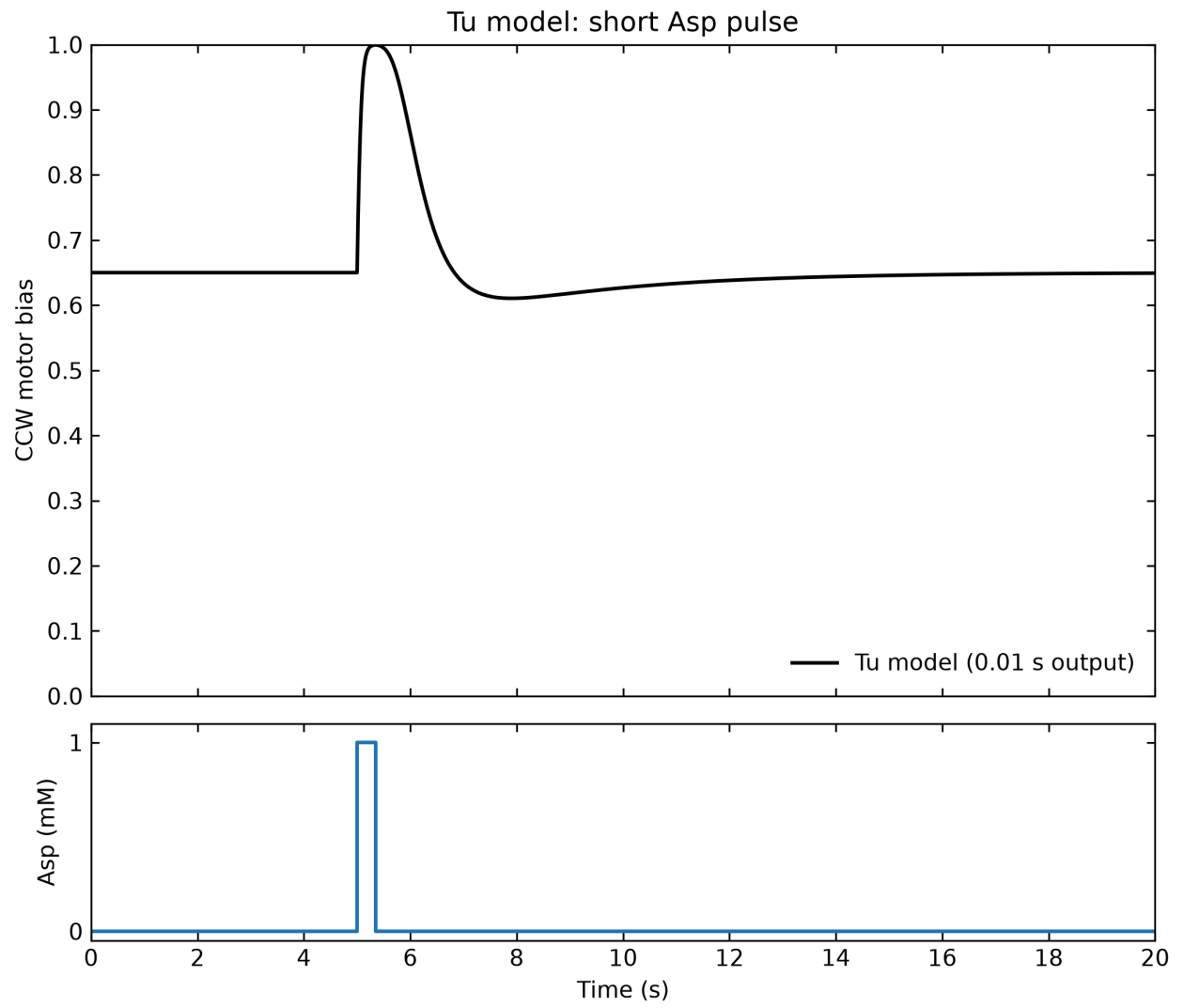
Analysis scheme

1. Impulse response: input aspartate concentration, measure CCW bias, CheY-P levels. Purpose: show it looks like a spike.
2. Step response: same as above. Purpose: show it doesn't look like a spike when the input is constantly on.
3. Single E coli trace: place E coli in a 1/2D environment, measure CCW bias and CheY-P levels. Purpose: show it performs chemotaxis, and show in reality CheY-P doesn't look like spikes (but if you consider $\text{CheY} - P$ to be fast positive feedback and m slow negative feedback, it can be similar to a spike; we can call it an "event" for now).
4. Population behavior: place multiple E coli on a 2D food source, observe if E coli population radiates outward like 4G7 video. Purpose: to show this is a working model by replicating the population behavior in biology. The parameters in the population vary, can relate to robustness (TODO: design this experiment).

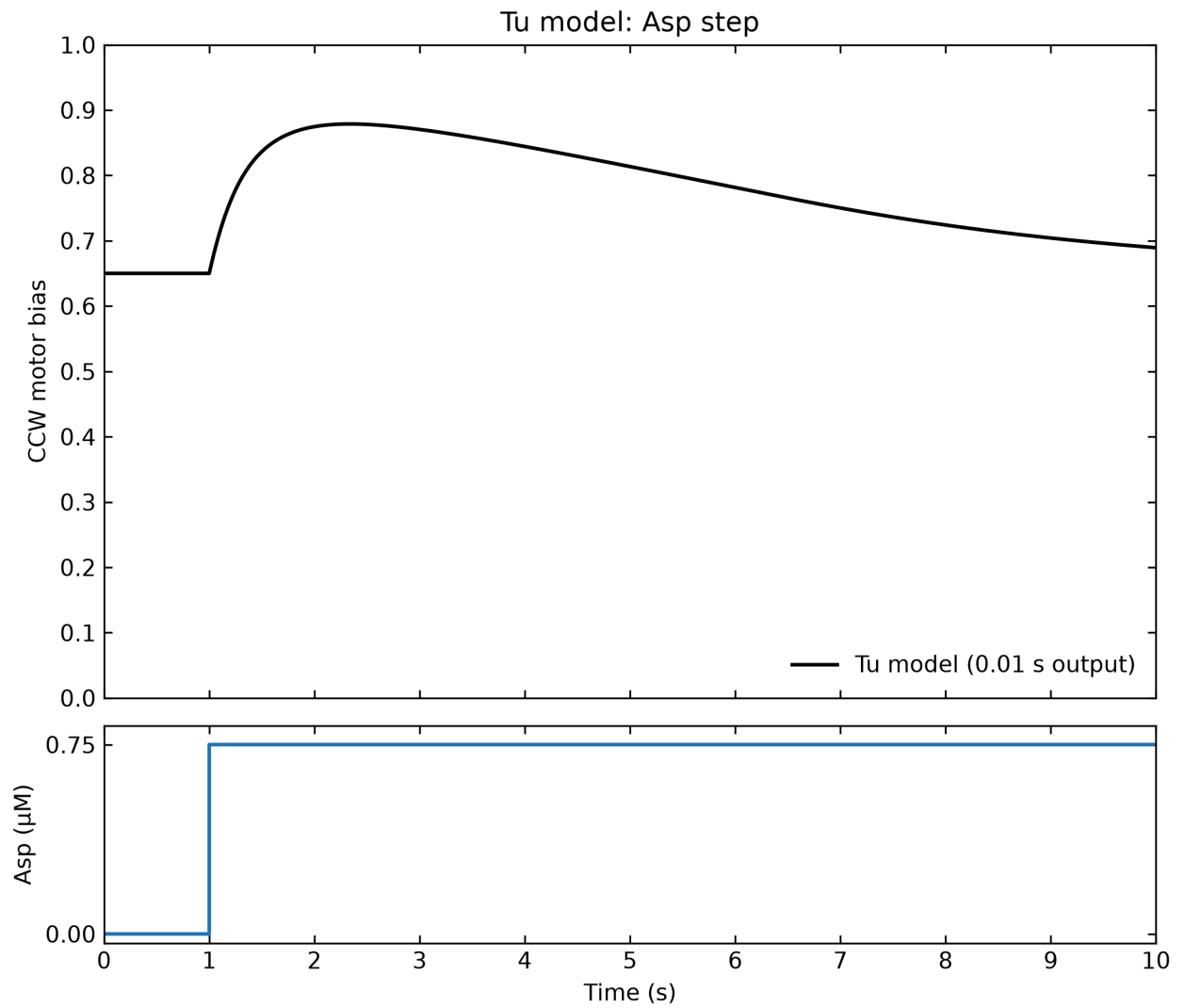
◆ Results

TODO: equations and results generated by Codex, to be verified.

1. Impulse response

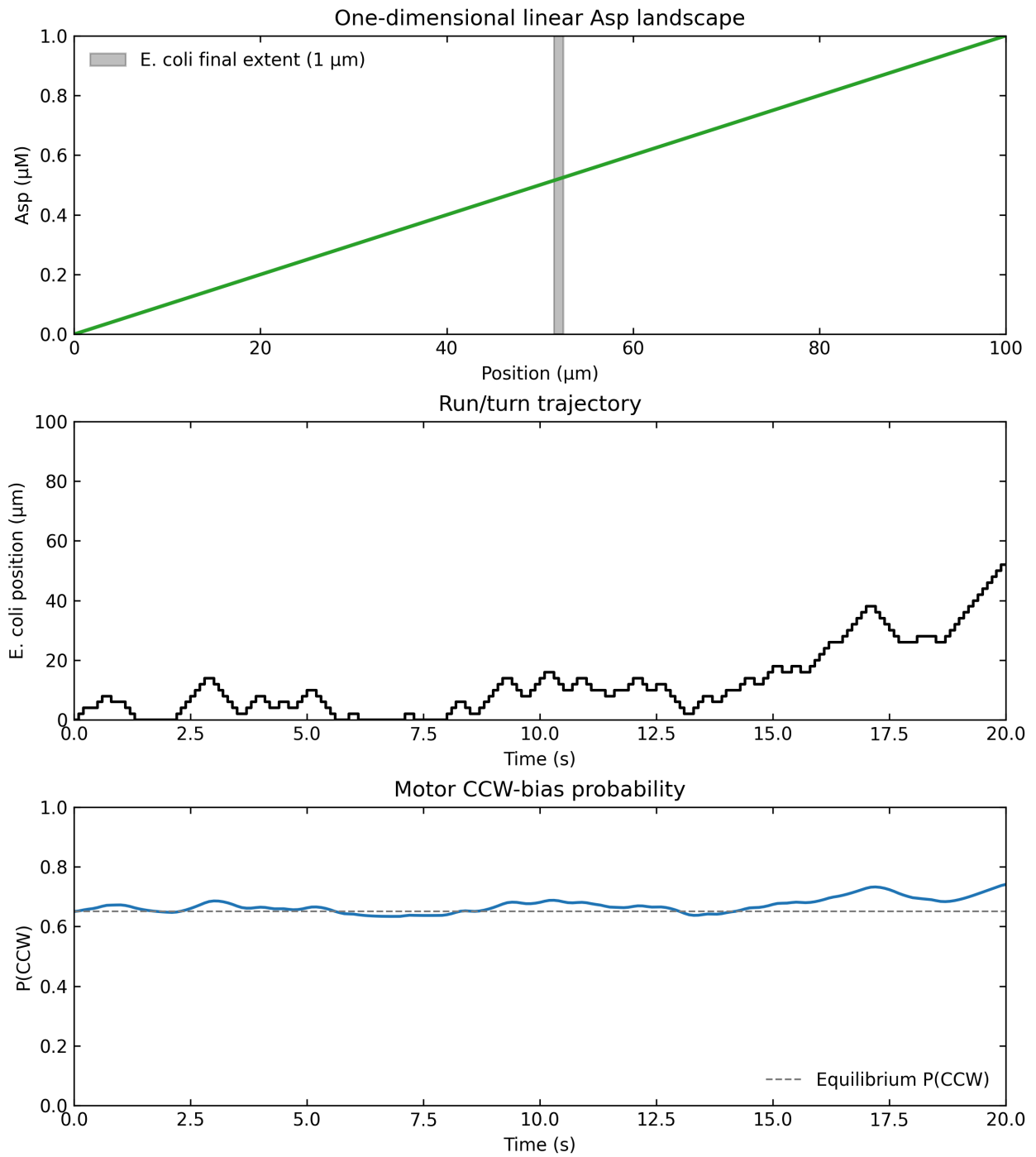


2. Step response

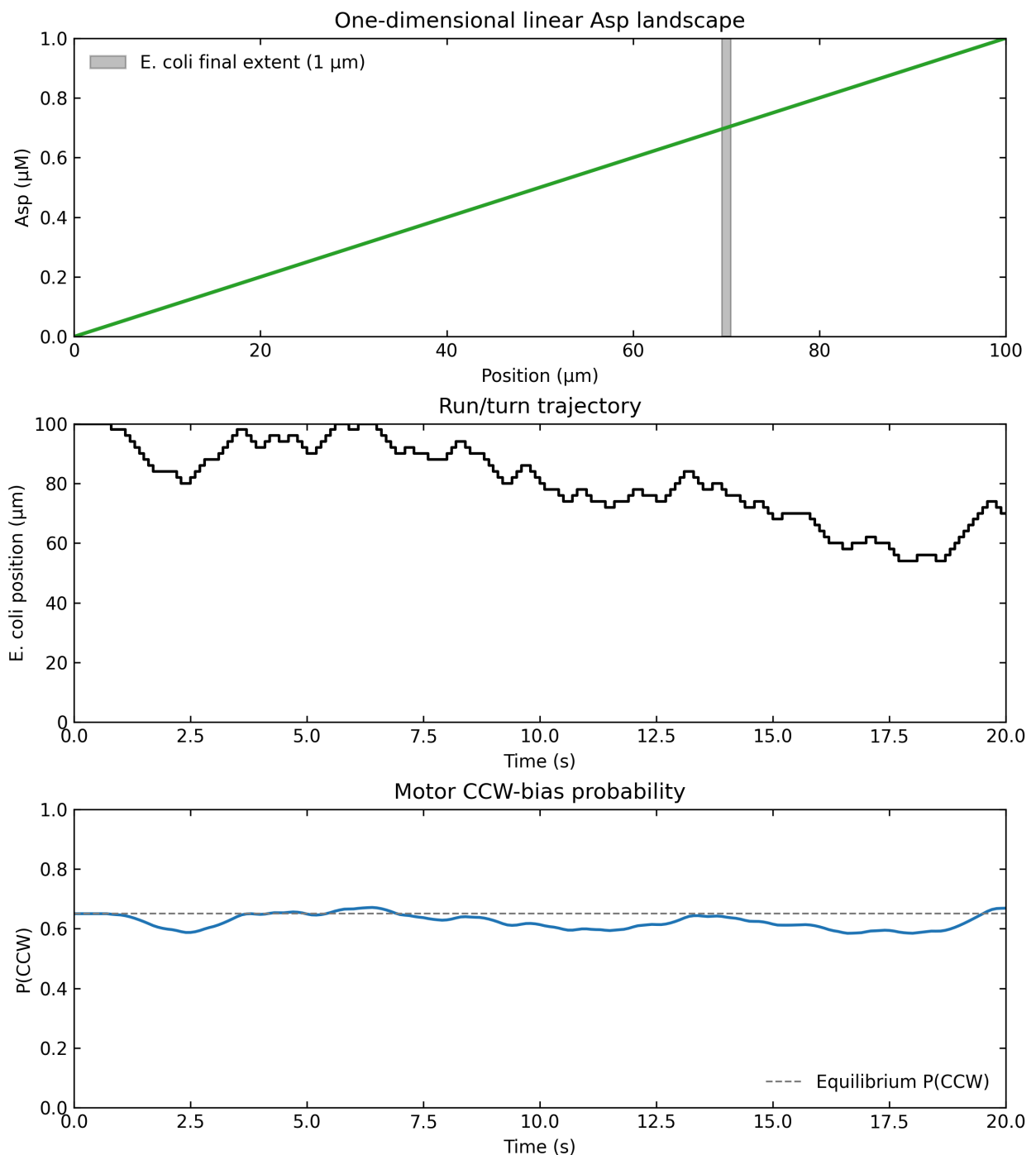


3. Single E coli trace in a 1D environment

If we initialize E coli at position=0 μm :



If we initialize E coli at position=100 μm :



2D environment: TODO

4. Population behavior: TODO

◆ Discussion

2. Spiking proprioception

Introduction

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Analysis scheme

🤔 To be resolved

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3. Spiking motor neuron activity

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4. Spike-based CPG controller

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 **To be resolved**

Results

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